Reading Orwell in Moscow

Natalia Vasilenok

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Overview

- Research Qustion: effect of conflict on individuals' willingness to seek **frames of reference(1)**, or heuristics that help individuals explain their social and political environment by means of analogy
    (1) **Frames of reference:** historical events people use to make sense of new events (e.g. understanding present Russian politics through past dictatorships)

- Setting: how Russia's full-scale invasion of Ukraine (contemporary political environment) in February 2022 reshaped readership (demand of frames of reference) of history and social science books in Russia

Overview of Empirical Analyses

- The Book Market Analysis (Supply Side)
  Test whether Russia's full-scale invasion of Ukraine (February 2022) changed the supply of history and social science books — specifically, whether newly published works devoted more attention to regime-critical topics.

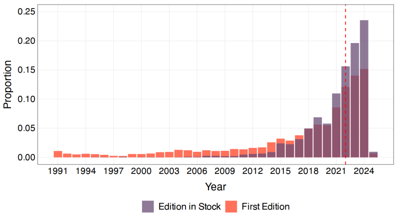
- The Readership Analysis (Demand Side)
  Test whether the invasion changed readers' preferences — i.e., whether people actually sought out those regime-critical frames of reference.

Book Market Analysis

- Identifying 4302 relevant books (what counts as a “frame of reference”)

1. Crawl the entire 125,000 online catalogue from Chitai-Gorod, Russia’s largest bookstore chain with abstract and metadata
2. Focus on titles under ”History and Society”
3. Exclude textbooks, educational materials, missing data and duplicates.

Figure E2: Books by Publication Year



Notes: The figure shows the distribution of books in the sample by publication year. The purple distribution represents the publication year of the edition in stock at *Chitai-Gorod* as of December 2024. The orange distribution represents the publication year of the first post-Soviet edition. Data are retrieved from the Russian State Library for 76% of the original sample and from LiveLib for 22%. The red dotted vertical line indicates the onset of Russia’s invasion of Ukraine.

Book Market Analysis

- Classifying books by their substantive content (Topic Modeling)

1. Preprocess to 4028 abstracts by filtering short abstracts

2. Fit STM with a Post-2021 covariate and topic correlations

3. Pick $K = 18$ by coherence and exclusivity

4. Label topics manually, validate them by Jaccard Similarity and cluster them by correlation

5. Assign books to topics via 50% rule

A Text Preprocessing

Preprocessing decisions can affect the results of unsupervised learning models, making it crucial to document and, whenever possible, substantiate all preprocessing steps (Denny and Spirling, 2018). Following prior literature, I remove punctuation, numbers, capitalization, and stop words. Similar to other languages, there is no unified and theoretically grounded list of stop words for Russian. Most lists would include prepositions, pronouns, numerals, and copular verbs. I use the list that contains 421 words and overlap it with the full list of prepositions in Russian language.[27] Additionally, I drop non-Cyrillic characters, single-character words, which are often abbreviations, and author names from each abstract. I then perform lemmatization by assigning inflected word forms to their canonical forms.

The language of book abstracts differs from other types of texts in its reliance on a specific vocabulary. First, abstracts aim to delineate a book's target audience. For example, they often state that 'a book is intended for a broad audience of readers.' This is reflected in the most frequent three-word combination in the corpus, 'broad audience reader', mentioned 345 times. Second, abstracts summarize the book's contents, which leads to overreliance on reporting language.[28] To capture abstract-specific language, I devise three strategies. First, from each abstract, I remove words that fall within the top 0.5% of the word frequency distribution. Across the entire corpus, the top three words are 'book' (used 5,142 times), 'history' (3,697 times), and 'war' (2,306 times). These words do not seem to provide substantive information about the books' contents, aside from indicating a focus on historical or societal topics, and can be considered stop words. Figure E3 in the Appendix shows 15 most frequent words along with their frequencies.

Second, I compile a list of abstract-specific words by comparing the corpus of book abstracts to the corpus of Russian mass media texts, which covers articles published on the online platforms of the 27 largest Russian mass media outlets between April 2016 and March 2017. I compute each word's relative frequency in the corpus of abstracts compared to the corpus of news and find the median of the frequency ratio distribution. I treat all words whose frequency ratio exceeds the median as abstract-specific. However, some words with high frequency ratios turn out to be infrequent in the corpus of abstracts, contradicting the definition of a stopword. I thus compute the 97.5% quantile of the abstract frequency distribution and exclude the words below this threshold from the list of abstract-specific words.

Third, I compute the frequencies of all three-word combinations, or 3-grams, that are formed after the exclusion of baseline stopwords, common words, and abstract-specific words. An examination of the most frequent 3-grams, such as 'contain comment hundred' or 'exert significant influence', suggests their mostly technical nature. Thus, I also drop words that enter 3-grams occurring more than four times in the corpus of abstracts.

As the final step, to reduce the sparsity of the document-term matrix and improve computational efficiency, I exclude all documents that consist of fewer than 15 words, which make up lower 5% of the document length distribution, and all words that appear in fewer than five documents. This leaves me with a sample of 4,028 documents.

For each document d:

1. **Draw topic proportions**

$$\theta_d \sim \mathrm{LogisticNormal}(X_d\gamma, \Sigma)$$

– θ_d: share of each topic in document d
– depends on covariates X_d (e.g., *Post-2021*)
– Σ: allows topics to be correlated

2. **For each word position n:**

$$z_{d,n} \sim \mathrm{Multinomial}(\theta_d), \qquad w_{d,n} \sim \mathrm{Multinomial}(\beta_{z_{d,n}})$$

– pick a topic $z_{d,n}$
– draw a word from that topic's word distribution β_k

Joint distribution:

$$P(w, z, \theta \mid X, \beta, \gamma, \Sigma) = \prod_d P(\theta_d \mid X_d, \gamma, \Sigma) \prod_n P(z_{d,n} \mid \theta_d) P(w_{d,n} \mid z_{d,n}, \beta)$$

Inference (reverse process):

$$P(\theta, z \mid w, X)$$

→ gives

- $P(\text{word} \mid \text{topic}) = \beta_k$ → **top words**
- $P(\text{topic} \mid \text{document}) = \theta_d$ → **topic mixture**

Structural Topic Modeling

STM is a generative probabilistic model that assumes every document is a mixture of topics, and every topic is a mixture of words.

Prior on Topic Proportion

Input: bag-of-words representation

Bayesian Inference

Output: $P(\text{topic} \mid \text{document})$

We have $\mathrm{P}(\text{topic}_i)$ *aggregated*

Book Market Analysis

- Structural Topic Modeling

Comparison with **SVD**$^{(1)}$ and **BERTopic**$^{(2)}$: model topic correlations

(1) **SVD:** singular value decomposition of a matrix, where rows are "word", columns are "document", entries are TF-IDF values.

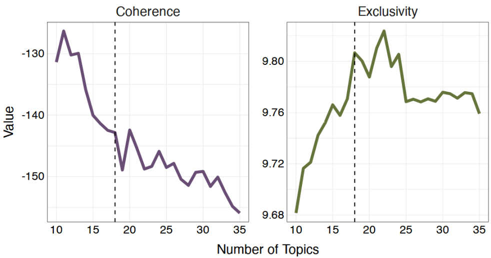
(2) **BERTopic:** transforms documents into dense BERT embeddings, reduces dimensions (e.g., UMAP), and clusters them (e.g., HDBSCAN) to form topics automatically, avoiding maunal labeling.

Question: Topic Prevelance is Post_2021-dependent, does this influence the magnificance of correlational estimate later?

Book Market Analysis

- Choose K = 18 by coherence and exclusivity
  Coherence: how often the top words of a topic appear together.
  → High coherence = words co-occur naturally (topic is meaningful).
  Exclusivity: how distinct each topic's top words are from other topics'.
  → High exclusivity = less word overlap between topics (topics are unique).

Figure E7: Choosing the Number of Topics

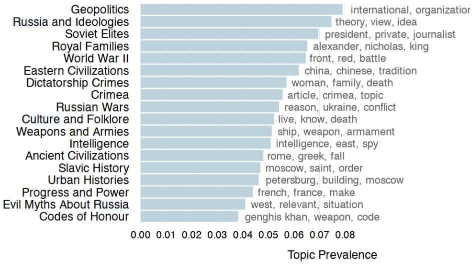


Notes: The figure shows the semantic coherence and exclusivity values for a correlated topic model fitted with a number of topics ranging from 10 to 35. The dashed black line indicates the chosen number of topics.

Book Market Analysis

- Manual Topic Labels and Clusters

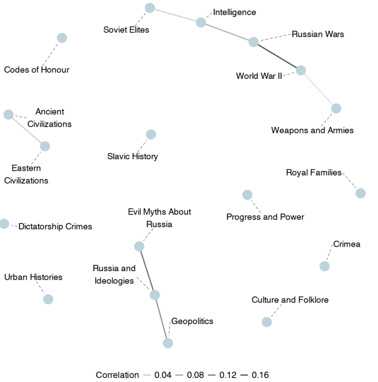
Figure 1: Book Topics in the *Chitay-Gorod* Catalogue



Notes: The figure shows the overall topic prevalence estimated from the correlated topic
18 topics, along with English translations of the three most frequent words associated with
Topics were manually labelled based on an in-depth reading of the most representative doc
the most frequent words associated with each topic.

Figure: Topics

Figure 2: Books on Dictatorship Crimes Are Unrelated to Books on World War II



Notes: The figure shows the estimated correlations among 18 topics fitted on the corpus of book abstracts. Correlations smaller than 0.01 have been set to zero.

Figure: Clusters

Representative Books by Cluster

Dictatorship Crimes (Regime-critical, Holocaust & complicity)

- *The Daughter of Auschwitz* (Tova Friedman & Malcolm Brabant).
- *The Broken House: Growing Up Under Hitler* (Horst Krüger).
- *The Conscience of a Nazi Judge* (Herlinde Pauer-Studer & J. David Velleman).

Ideology (Russian exceptionalism & anti-Western rhetoric)

- *Evil Myths About Russia: What the West Says About Us*.

Book Market Analysis

- Effect of Invasion on Topic Distribution in Publishment

Table 1: Topic Prevalence and Russia's Invasion of Ukraine

	Dependent variable:			
	Topic Prevalence, %			
	Dictatorship Crimes (1)	Ideology (2)	Russian Wars (3)	Other (4)
Panel A: No Controls				
Published after 2021	1.809***	−3.404**	0.379	1.216
	(0.645)	(1.48)	(2.681)	(2.966)
Mean of DV	5.71	19.46	29.04	45.8
Standard deviation of DV	14.41	25.58	32.4	35.46
Observations	4,028	4,028	4,028	4,028
Panel B: Controls				
Published after 2021	1.729**	−1.956	−2.187	2.413
	(0.732)	(1.423)	(2.106)	(2.778)
Price, in 100 Rubles	−0.116**	−0.184*	0.193	0.107
	(0.045)	(0.094)	(0.187)	(0.17)
Copies, in 100 Copies	0.001	0.092**	−0.087	−0.006
	(0.028)	(0.042)	(0.073)	(0.041)
Popularity, in 100 Readers	0.173**	−0.106	−0.194**	0.128
	(0.081)	(0.099)	(0.062)	(0.055)
Average Rating	0.706***	−0.618**	−2.005***	1.917***
	(0.199)	(0.225)	(0.438)	(0.468)
Publisher Fixed Effects	✓	✓	✓	✓
Mean of DV	6.17	18.83	29.45	45.55
Standard deviation of DV	15.08	25.22	32.59	35.59
Observations	3,239	3,239	3,239	3,239

Notes: The unit of observation is a book. The dependent variable is the sum of topic prevalence estimates derived from a structural topic model for a given topic cluster, scaled between 0 and 100. Standard errors, computed using the posterior distribution of the topic prevalence estimates and adjusted for clustering at the publisher level, are shown in parentheses.

$$Pr(Topic)_i = \beta_0 + \beta_1 Post\ 2021_i + XB_i + \mu_j + \varepsilon_i$$

DV: topic prevalence from STM estimates, scaled between 0-100

Panel A: full sample (4,028 abstracts).

Panel B: reduced sample (3,239) due to missing controls.

Post 2021: yearly dummy = 1 if book published $\geq$ 2022; metadata contain year only, not month.

Book Readership Analysis

- Focus on "Dictatorship Crime" Topic

$$Read_{ikt} = \beta_1 Post\ Invasion_t \times Topic_{ik} + \mu_{ik} + \eta_t + \varepsilon_{ikt} \tag{2}$$

Unit of observation: user–date–topic

Variable definitions:

$Read_{ikt}$: share of topic-k-book relative to all books that user i read in month t.

$Post\ Invasion_t$: dummy $= 1$ for March 2022 onward.

$Topic_{ik}$: dummy $= 1$ for Dictatorship Crimes and 0 for comparison topics.

β_1: *do not capture direction of change. To fix it, estimate separate event-study models without fixed time effect because they actually subtract average DV in this month.*

Book Readership Analysis

- Comparison Group: Ancient Civilization because:

  1) similar pre-treatment characteristics.

  2) not likely to exhibit systematic response to invasion.

  Ideology and War also used for robustness
- Question: No Response?

Table D4: Choosing the Comparison Topic

	Books (1)	Readers, 2018-22 (2)	Readers, Total (3)	Price, Rubles (4)	Copies (5)
Clusters					
Dictatorship Crimes	126	38.7	191.1	903.6	2,201.4
Ideology	557	7.5	27.9	952.1	1,697.0
Russian Wars	1,039	2.2	8.5	1,068.2	1,315.8
Topics					
Ancient Civilizations	120	31.0	100.7	929.1	1,914.4
Codes of Honour	87	11.5	80.8	940.2	2,039.5
Crimea	104	8.3	35.4	1,286.1	1,470.8
Culture and Folklore	90	16.9	87.4	913.9	2,634.8
Eastern Civilizations	150	18.3	89.9	1,167.5	1,521.4
Progress and Power	89	10.0	44.0	985.8	2,367.7
Royal Families	161	6.1	25.3	916.2	1,934.0
Slavic History	125	13.7	16.6	951.0	1,546.3
Urban Histories	117	8.3	28.5	1,249.3	1,835.3

Notes: The unit of observation is a book, whose abstract contains more than 50% of a given topic. Columns (2) and (3) report data from LiveLib: the average number of readers between January 2018 and January 2022, and over the entire period of the platform's operation. Column (4) reports the average price in rubles, and Column (5) reports the number of printed copies of the edition in stock at *Chitay-Gorod*.

Book Readership Analysis

1. Period of analysis

Monthly panel data from **January 2018 to May 2025**.

2. Books

Subset of **4,028 books** from the *Chitai-Gorod* catalogue.
Treatment: **Dictatorship Crimes** topic (126).
Comparison: **Ancient Civilizations** (120), **Ideology** (557), and **Russian Wars** (1039).

3. Readers

Initial sample: **105,215 users** on the *LiveLib* reading platform.
After filtering for users who registered and logged after 2018, and logged after March 2023, and removing users who logged lifetime reading on the registration month, the final sample includes **26,635 active readers.**

But it's not representative of Russian population

Book Readership Analysis

Table D2: Descriptive Statistics

Variable	Mean	St.Dev.	Median	Min	Max	N
Panel A: Books Sample						
Words in Abstract	45.4	24.0	42.0	1	257	4,302
Published after 2021	0.6	0.5	1.0	0	1	4,302
Copies Printed	1,694.5	1,915.1	1,200.0	20	40,002	3,770
Price (Rubles)	1,027.3	729.7	839.0	87	9,870	4,001
Reprinted after 2021	0.2	0.4	0.0	0	1	4,270
Users Read	43.9	321.9	2.0	0	11,144	4,201
Users Planning	79.9	346.8	2.0	0	8,312	4,201
Average Rating	2.4	2.0	3.5	0	5	4,201
Users Rated	42.2	311.1	2.0	0	10,920	4,201
Panel B: Readers Sample						
Registered before January 1, 2018	0.4	0.5	0	0	1	95,913
Registered after February 24, 2022	0.2	0.4	0	0	1	95,913
Active until February 24, 2023	0.7	0.4	0	0	1	95,913
Books Logged, Per Month	1.8	10.0	0	0	10,369	8,153,030
History Books Logged, Per Month	0	0.4	0	0	261	8,153,030

Book Readership Analysis

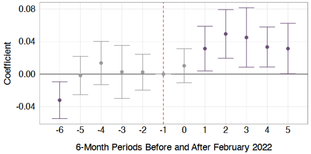
Table 2: Changes in 'Dictatorship Crimes' Readership Rates and Russia's Invasion of Ukraine

		Dependent variable:		
		Read, %		
	Reference	(1)	(2)	(3)
Panel A				
Post Invasion × Dictatorship Crimes	Ancient Civilizations	0.050***	0.050***	0.043***
		(0.007)	(0.007)	(0.007)
Panel B				
Post Invasion × Dictatorship Crimes	Ideology	0.033***	0.033***	0.032***
		(0.006)	(0.006)	(0.008)
Panel C				
Post Invasion × Dictatorship Crimes	Russian Wars	0.033***	0.034***	0.021***
		(0.007)	(0.007)	(0.008)
User-Topic Fixed Effects		✓	✓	✓
Month-Year Fixed Effects		✓	✓	✓
Pre-Invasion Treated Mean		0.061	0.062	0.063
Pre-Invasion Treated SD		1.851	1.851	1.864
Geographic Sample		All Users	Not Ukraine	Russia
Unique Users		26,635	25,671	14,528
Observations		2,781,968	2,684,064	1,578,556

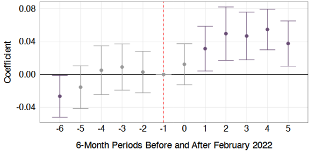
Notes: The unit of observation is a user-date-topic triple. The dependent variable is a percentage of books under topic k — defined as books whose abstracts are more than 50% comprised of the respective topic as defined by the structural topic model — relative to the total number of books recorded by the same user in month t. The panel covers the period from January 2018 to May 2025. Standard errors, clustered by user and month, reported in parentheses.

Book Readership Analysis

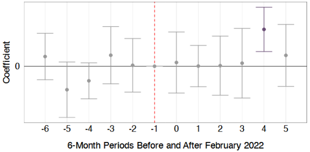
Figure 4: Readership of 'Dictatorship Crimes' Books Rises After the Invasion



(a) Changes in 'Dictatorship Crimes' Readership Rates Relative to 'Ancient Civilizations' Readership Rates



(b) Changes in 'Dictatorship Crimes' Readership Rates



(c) Changes in 'Ancient Civilizations' Readership Rates

Full Sample of active users (95913?)

Robust to non-Ukraine users (90000?)

Indicate both the relative difference and the direction of change

Lead to an increase in domestic interest in war-related topics, espeacially for books on the Holocaust, Nazi Germany, and everyday life under dictatorship.

Interpret as evidence that dictatorship topic prevalence draws an analogy between the experience of living in contemporary Russia and that of living under past dictatorial regimes.

Discussion

David *2025-09-25 03:11:39* 35 -3 Hide Replies #

How do we know this isn't from Russians associating dictatorship crimes to Ukraine, rather than a way to show dissent via book buying?

It seems like the Russian authorities often suggested that Ukraine was like Nazi Germany.

> *rayward* *2025-09-25 05:35:22* 12 -4 Hide Replies #
>
> Putin called the Ukrainian government and the Ukrainians "Neo-Nazis" and "fascists", and described the invasion of Ukraine as a "special military operation" for the "demilitarizations" and "denazification" of Ukraine. Thus, the book purchasing in Russia after the invasion could be interpreted more as support for than dissent against the invasion. One should recall that the Soviet Union signed a pact with Nazi Germany (the ""Nazi-Soviet Pact"), a no-aggression treaty signed in 1939 that included secret provisions to partition Eastern Europe including Poland, and gave the green light to Germany to invade Poland unopposed and gave Germany time to build its military. Alas, Germany invaded the Soviet Union in June 1941. Nazi Germany and the Soviet Union deserved each other. Young people today seem to know little about history.
>
> > *Eric Mack* *2025-09-25 09:06:54* 6 0 #
> >
> > I agree totally. It is (at first) astonishing that the author interprets the increased purchase of books on Nazi Germany as evidence of increased disfavor of the Putin regime. My question is, Why does "Orwell" figure in the title of the entry? If one goes beyond the abstract of the paper, does the author reveal a remarkable increase in the sale of 1984? Now that would be interesting.

Discussion

- We know in the supply side Dictatorship Topic has risen a lot
  Is it possible the rise in Readership is due to richer supply instead of frames of reference?

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Observations	[illegible]	[illegible]	[illegible]	[illegible]